Sequences and Functions 2

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1 Functions

1.1 Power Series

Definition. A **power series** is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n.$$

where $a_0, a_1, \ldots$ are all real numbers, called the coefficients of the power series.

Theorem. For any power series $\sum_{n=0}^{\infty} a_n x^n$ either

1. $\sum_{n=0}^{\infty} |a_n x^n|$ converges $\forall x \in \mathbb{R}$
2. $\sum_{n=0}^{\infty} |a_n x^n|$ diverges $\forall x \in \mathbb{R} \setminus \{0\}$
3. $\exists R \in (0, \infty)$ such that $\sum_{n=0}^{\infty} |a_n x^n|$ converges if $|x| < R$ and diverges if $|x| > R$.

Proof. Suppose their exists at least one $x > 0$ such that the power series converges and at least one $z > 0$ such that it doesn't.

Define $S := \{r \geq 0 \mid (|a_n| r^n)_{n \in \mathbb{N}_0} \text{ is bounded}\}$ and $R := \sup S$. Hence $x \in S$. Suppose there exists $r > z \in S$. Then

$$|a_n z^n| = \frac{|a_n| z^n}{|a_n| r^n} |a_n| r^n = (|a_n| r^n) \left(\frac{z^n}{r^n}\right) \leq M \left(\frac{z}{r}\right)^n$$

for some M. Since this is less than a geometric sequence, the sum must converge, a contradiction. This means z is an upper bound of S.

For any $x \in S$, we can choose a $x < r < R$. Then by the same argument as above, $\sum_{n=1}^{\infty} |a_n x^n|$ is less than a geometric series, so converges. If $x > R$, then $a_n x^n$ is unbounded and clearly does not converge.

$\square$

Remark. In Case 3, the sequence may or may not converge if $|x| = R$, and it depends on the sequence $(a_n)_{n \in \mathbb{N}}$.

Definition. (Radius of Convergence)

- If $\sum_{n=0}^{\infty} |a_n x^n|$ converges $\forall x \in \mathbb{R}$, we define the **radius of convergence** to be ∞
- If $\sum_{n=0}^{\infty} |a_n x^n|$ diverges for all $|x| > 0$ we define the radius of convergence to be 0.
- If $\sum_{n=0}^{\infty} |a_n x^n|$ converges $|x| < R$ and diverges $|x| > R$, then the **radius of convergence** is defined as R.

Although we know the radius of convergence exists, we know nothing about it. The next theorem gives us some information about it.

Theorem. For any power series $\sum_{n=0}^{\infty} |a_n x^n|$, if

- $a_n \neq 0 \quad \forall n \in \mathbb{N}_0$ and $R = \lim_{n \to \infty} \frac{|a_n|}{|a_{n+1}|}$ exists, then R is the radius of convergence.

Proof. By the ratio test

$$\frac{a_{n+1} x^{n+1}}{a_n x^n} = x \left(\frac{a_{n+1}}{a_n}\right)$$

If $R = \infty$, this always converges to 0, so the power series is convergent for all x. Conversely, if $R = 0$, the ratio test gives $\infty > 1$ for all x. If $R = R$, then

$$\begin{aligned} \lim_{n\to\infty} x\left(\frac{a_{n+1}}{a_n}\right) &< 1 \\ \frac{x}{R} &< 1 \\ x &< R. \end{aligned}$$

□

1.2 Exponential and Trigonometic Functions

A power series with radius of convergence $R \in (0, \infty]$ gives a function $f : (-R, R) \to \mathbb{R}$. Many important functions can be defined in this way.

Definition. For $x \in \mathbb{R}$, define

$$\exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Proposition. The power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ has infinite radius of convergence, thus the exponential function has domain equal to $\mathbb{R}$.

Proposition.

$$\exp(x + y) = \exp(x)\exp(y).$$

Proof. Non-examinable. □

Corollary. For any $x \in \mathbb{R}$,

- $\exp(x) > 0$
- $\exp(-x) = \frac{1}{\exp(x)}$.

Proof.

- We know $\exp(0) = 1$. Then

$$\begin{aligned} \exp(x - x) &= 1 \\ \exp(x)\exp(-x) &= 1\exp(-x) \qquad = \frac{1}{\exp(x)}. \end{aligned}$$

- If $x \geq 0$, $\exp(x) \geq 1 > 0$. Then

$$\exp(-x) = \frac{1}{\exp(x)} > 0.$$

□

Definition. For all $x \in \mathbb{R}$,

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

Remark. We cannot find the radius of convergence using the ratio test, as many terms are equal to 0. However, we can compare these series with the exponential function and show they are bounded by the comparison test.

1.3 Limits of Functions

Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$.

Definition. Let $S \subseteq \mathbb{R}$ and $c \in \mathbb{R}$. We say c is a **cluster point** of S if there exists $(x_n)_{n\in\mathbb{N}}$ in $S \setminus \{c\}$ such that $\lim_{n\to\infty} x_n = c$.

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Let $L \in \mathbb{R}$ and let c be a cluster point of D.

We say f **converges to** L as x tends to c, or $\lim_{x\to c} f(x) = L$ if

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall x \in D \quad \text{s.t.}$$

$$\text{if } 0 < |x - c| < \delta \text{ then } |f(x) - L| < \epsilon$$

In the above definition, the definition of $f(c)$ is irrelevent.

Theorem. Let $c \in \mathbb{R}$ be a cluster point of $D \subseteq \mathbb{R}$. Let $f : D \to \mathbb{R}$ and $L \in \mathbb{R}$. Then the following are equivalent:

- $\lim_{x\to c} f(x) = L$
- For any sequence $(x_n)_{n\in\mathbb{N}}$ in $D \setminus \{c\}$

$$\lim_{n\to\infty} x_n = c \Rightarrow \lim_{n\to\infty} f(x_n) = L.$$

Proof.

- Suppose $\lim_{x\to c} f(x) = L$ and $\lim_{n\to\infty} x_n = c$. Now fix $\epsilon > 0$. Then we obtain a δ where

$$0 < |x - c| < \delta \Rightarrow |f(x) - L| < \epsilon.$$

Using this δ, we obtain an N such that

$$\forall n > N \quad |x_n - c| < \delta.$$

Then since $x_n \neq c$ for all x, the conditions for $f(x_n) \to L$ are met.

- Since assuming an implication is difficult to work with, we instead prove the contrapositive:

$$\lim_{x \to c} f(x) \neq L \wedge x_n \to c \Rightarrow \forall \epsilon > 0,\ \exists N,\ \forall n \geq N \quad |f(x_n) - L| > \epsilon.$$

The negation of the limit of f gives us an ϵ where for any δ we can find an x such that $0 < |x - c| < \delta$ and $|f(x) - L| > \epsilon$. Then, using this ϵ, fix $\delta > 0$. Choose N such that

$$|x - c| < \delta$$

and then the conclusion follows.

□

Proposition. Let c be a cluster point of $D \subseteq \mathbb{R}$. Suppose there exists a sequence $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ in $D \setminus \{c\}$ such that

- $\lim_{n \to \infty} x_n = c$ and $\lim_{y \to \infty} = c$
- $\lim_{n \to \infty} f(x_n) = A$ and $\lim_{n \to \infty} f(y_n) = B$ with $A \neq B$.

Then we conclude $\lim_{x \to c} f(x)$ cannot exist.

Proof. Suppose the limit exists. Then since x_n and y_n have the same limit

$$\lim_{n \to \infty} f(x_n) = A = B = \lim_{n \to \infty} f(y_n)$$

which is a contradiction. □

Proposition. Let c be a cluster point of $D \subseteq \mathbb{R}$ and let $f : D \to \mathbb{R}$. If $\lim_{x \to c} f(x) = L_1$ and $\lim_{x \to c} f(x) = L_2$ then

$$L_1 = L_2.$$

Proof. $\lim_{x \to c} f(x) = L$ is equivalent to $f(x_n) \to L$ for $x_n \to c$. Then

$$\lim_{n \to \infty} f(x_n) = L_1 = L_2.$$

So $L_1 = L_2$ by uniqueness of limits. □

Theorem. (Algebra of limits for functions)

Suppose $D \subseteq \mathbb{R}$ and c is a cluster point of D. Let $f, g : D \to \mathbb{R}$ be two functions.

Suppose $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} g(x) = M$.

Let $x \in \mathbb{R}$. Then

- $\lim_{x \to c} (f(x) + g(x)) = L + M$
- $\lim_{x \to c} \alpha f(x) = \alpha L$
- $\lim_{x \to c} f(x)g(x) = LM$
- If $M \neq 0$, then $\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{L}{M}$

Proof. By turning the limit of the function into a sequence limit, they can be easily proved using the algebra of limits. $\square$

Theorem. Suppose $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ is a cluster point of D, $f : D \to \mathbb{R}$ and $\lim_{x \to c} f(x) = L$. Then

- If $f(x) \leq M \quad \forall x \in D$, then $L \leq M$
- If $f(x) \geq M \quad \forall x \in D$, then $L \geq M$.

Proof. Consider WLOG $f(x) \leq M$. Then for $x_n \to \infty$

$$\lim_{n \to \infty} f(x_n) \leq M$$

so since limits preserve non-strict inequalities, $L \leq M$. $\square$

Theorem. Suppose $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ is a cluster point of D and $f, g, h : D \to \mathbb{R}$ with

$$f(x) \leq g(x) \leq h(x) \quad \forall x \in D.$$

and

$$\lim_{x \to c} f(x) = L = \lim_{x \to c} h(x).$$

Then $\lim_{x \to c} g(x) = L$.

Proof. Consider $x_n \to c$. Then

$$f(x_n) \leq g(x_n) \leq h(x_n)$$
$$\lim_{n \to \infty} f(x_n) \leq \lim_{n \to \infty} g(x_n) \leq \lim_{n \to \infty} h(x_n).$$

so by sandwhich theorem the limit of g at c is L. $\square$

Now we will consider the limit of a function to ∞.

Definition. Let $D \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $L \in \mathbb{R}$. Suppose there exists a sequence $(a_n)_{n\in\mathbb{N}}$ in D such that $\lim_{n\to\infty} a_n = \infty$.

We say f converges to L as $x \to \infty$ if

$$\forall \epsilon > 0 \quad \exists N \in \mathbb{R} \text{ s.t. } \forall x \in D, \quad x > N$$

$$|f(x) - L| < \epsilon.$$

Hence we write $\lim_{x\to\infty} f(x) = L$.

Definition. Let $D \subseteq \mathbb{R}$, $c \in \mathbb{R}$ be a cluster point of D and $f : D \to \mathbb{R}$

- We say f diverges to ∞ as $x \to c$ if $\forall M \in \mathbb{R}\ \exists \delta > 0$ such that $\forall x \in D$ with $0 < |x - c| < \delta$ we have $f(x) > M$. In this case, we write $\lim_{x\to c} f(x) = \infty$.
- We say f diverges to $-\infty$ as $x \to c$ if $\forall M \in \mathbb{R}\ \exists \delta > 0$ such that $\forall x \in D$ with $0 < |x - c| < \delta$ we have $f(x) < M$. In this case, we write $\lim_{x\to c} f(x) = -\infty$.

2 Continuity

Definition. Let $D \subseteq \mathbb{R}$, $c \in D$ and $f : D \to \mathbb{R}$. We say f is continuous at c if

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall x \in D, \quad |x - c| < \delta$$

$$|f(x) - f(c)| < \epsilon.$$

We say f is **continuous** on D if this holds true at every $c \in D$.

Theorem. Let $D \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and c be a cluster point of D.

Then f is continuous at c if and only if

$$\lim_{x\to c} f(x) = f(c).$$

Proof. Suppose f is continuous at c. Fix $\epsilon > 0$ and obtain δ from continuity of f. Then for all x with $0 < |x - c| < \delta$, we have

$$|f(x) - f(c)| < \epsilon.$$

Thus f converges to $f(c)$ at ϵ. Now suppose $\lim_{x\to c} f(x) = f(c)$. Fix $\epsilon > 0$ and obtain δ from the limit of f. Then for all $0 < |x - c| < \delta$,

$$|f(x) - f(c)| < \epsilon.$$

For the case where $|x - c| = 0$, $x = c$ and $|f(c) - f(c)| = 0 < \delta$, so f is continuous. $\square$

Remark. The relation between continuity at c and $\lim_{x \to c} f(x) = f(c)$ does not hold if c is not a cluster point. However, functions are always continuous at points that are not cluster points of the domain.

Theorem. (Algebra of continuous functions)
Let $D \subseteq \mathbb{R}$, $c \in D$ and $\alpha \in \mathbb{R}$. Let $f, g : D \to \mathbb{R}$ be continuous at c, then

- $f + g$
- αf
- fg

are all continuous at c.
If $g(c) \neq 0$, then $\frac{f}{g}$ is also continuous at c.

Proof. Use the algebra of limits. □

2.1 Composition of Continuous Functions

Given two sets $D, E \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $g : E \to D$, we can define the composite function

$$(f \circ g)(x) := f(g(x)).$$

Theorem. (Continuity of Continuous Functions)
Let $D, E \subseteq \mathbb{R}$, and $c \in E$. Let $f : D \to \mathbb{R}$ and $g : E \to D$.
If g is continuous at c and f is continuous at $g(c)$, then $f \circ g$ is continuous at c.

Proof. Let $\epsilon > 0$. Because f is continuous at $g(c)$, there exists $\rho > 0$ such that

$$|y - g(c)| < \rho \Rightarrow |f(y) - f(g(c))| < \epsilon.$$

Then since g is continuous at c we can find a δ such that

$$|x - c| < \delta \Rightarrow |g(x) - g(c)| < \rho.$$

□

2.2 Continuity of Selected Functions

Proposition. The following functions are all continuous.

- $f(x) = \alpha$ for $\alpha \in \mathbb{R}$.
- $f(x) = x$
- $f(x) = \sum_{i=n}^{\infty} a_n x^n$

- $f(x) = \exp(x)$
- $f(x) = \sinh(x)$
- $f(x) = \cosh(x)$
- $f(x) = \sin(x)$
- $f(x) = \cos(x)$
- $f(x) = \sec(x), g(x) = \tan(x)$ where $f, g : \mathbb{R} \setminus \{\frac{\pi}{2} + n\pi | n \in \mathbb{Z}\} \to \mathbb{R}$
- $f(x) = \csc(x), g(x) = \cot(x)$ where $f, g : \mathbb{R} \setminus \{n\pi | n \in \mathbb{Z}\} \to \mathbb{R}$

Proof.

- Constant and identity functions are trivial, and from these polynomials can be built.
- For $f(x) = \exp(x)$, we first prove $\exp(x)$ is continuous at 0. Let $\epsilon > 0$ and

$$\begin{aligned} |\exp(x) - \exp(0)| &= |\exp(x) - 1| \\ &= |1 + \sum_{i=1}^{\infty} \frac{x^n}{n!} - 1| \quad =\leq \gamma|x| \quad \text{(non-examinable)}. \end{aligned}$$

  Thus if $|x| < \delta = \frac{\epsilon}{\gamma}$ then $\lim_{x \to 0} \exp(x) = 1$.

  Now for an arbitrary c, then

$$\exp(x) = \exp(x - c)\exp(c)$$

  so the first term converges to 0 which is continuous, and the second is constant thus exp is continuous on $\mathbb{R}$. This also gives continuity for sinh and cosh.

- The same argument can prove that cos and sin are continuous at 0. Then for any c,

$$\begin{aligned} \cos(x) &= \cos(x - c + c) = \cos(x - c)\cos(c) - \sin(x - c)\sin c \\ \sin(x) &= \sin(x - c + c) = \sin(x - c)\cos(c) - \cos(x - c)\sin(c) \end{aligned}$$

  which are both continuous also by the same argument. This also gives continuity for $\tan, \sec$ etc.

□

2.3 The Intermediatve Value Theorem

Let $D \subseteq \mathbb{R}$, $\alpha \in \mathbb{R}$ and $f : D \to \mathbb{R}$. Solving the equation $f(x) = \alpha$ can rarely be done explicitly. However, if f is continuous on $D = [a, b]$, and if α is between $f(a)$ and $f(b)$, it is plausible that $f(x) = \alpha$ has a solution in D.

Theorem. Let $a, b \in \mathbb{R}$ with $a < b$. Suppose

- $f : [a, b] \to \mathbb{R}$ is continuous
- $f(a) \leq \alpha \leq f(b)$ or $f(b) \leq \alpha \leq f(a)$.

Then there exists $c \in [a, b]$ such that

$$f(c) = \alpha.$$

Proof. We want to construct sequences $(x_n)_{n\in\mathbb{N}}$ $(y_n)_{n\in\mathbb{N}}$ that both converge to c with $f(x_n) \leq \alpha$ and $f(y_n) \geq \alpha$. Then

$$f(c) = \lim_{n\to\infty} f(x_n) \leq \alpha \leq \lim_{n\to\infty} f(y_n) = f(c).$$

We construct these sequences by setting $x_0 = a$ and $y_0 = b$. Then we calculate z_0, the midpoint, and if $f(z_0) \leq \alpha$, we set $x_1 = x_0$ and $y_1 = y_0$, and otherwise do it the other way around. This can be generalized for any n, giving a sequence with the desired properties. □

2.4 The Weierstrass Extreme Value Theorem

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$.

- We say $c \in D$ is a **minimum point** of f in D if

$$f(c) \leq f(x) \quad \forall x \in D.$$

- We say $c \in D$ is a **maximum point** of f in D if

$$f(c) \geq f(x) \quad \forall x \in D.$$

We say f **attains** its maximum or minimum in D if a maximum or minimum point exists in D.

Theorem. Let $a, b \in \mathbb{R}$ with $a < b$. Suppose $f : [a, b] \to \mathbb{R}$ is continuous. Then

- there exists finite $L > 0$ such that

$$|f(x)| \leq L \quad \forall x \in [a, b].$$

- f attains its maximum and minimum in $[a, b]$.

Proof.

- By contradiction, we have for any n, there exists an $x \in [a, b]$ such that

$$f(x) > n.$$

Define $(x_n)_{n\in\mathbb{N}}$ in this way. Then this sequence is clearly bounded so has a convergent subsequence $x_{n_k} \to c$. As f is continuous

$$f(c) = \lim_{n\to\infty} f(x_{n_k}) = \infty$$

which is a contradiction as $f(c)$ is clearly bounded.

- Let $M = \sup\{f(x)|x \in [a,b]\}$ which is bounded. Then for all n there exists a y such that

$$M - \frac{1}{n} \leq y \leq M$$

and define the sequence y_n in this way, so $y_n \to M$. Since y is bounded it has a convergent subsequence $y_{n_k} \to c \in [a,b]$ and also by continuity

$$f(c) = \lim_{n\to\infty} f(y_{n_k}) = M.$$

hence f obtains a maximum. The same argument can be used for the supremum.

□

2.5 The Natural Logarithm

Before we can define the inverse of the exponential function, we must show that it is a bijection.

Proposition. The exponential function is bijective.

Proof. We first prove injectivity and then surjectivity.

- Let $x, y \in \mathbb{R}$ with $x > y$. Then

$$\begin{aligned}\frac{\exp(x)}{\exp(y)} &= \exp(x-y) \\ &= 1 + \sum_{n=1}^{\infty} \frac{(x-y)^n}{n!} > 1.\end{aligned}$$

Therefore, as the sum is positive, $\exp(x) > \exp(y)$. This implies that $\exp : \mathbb{R} \to (0,\infty)$ is injective.

- Fix $y > 0$. Then

$$\exp(y) = 1 + y + \sum_{n=2}^{\infty} \frac{y^n}{n!} > y.$$

This implies, for $y > 0$, $\exp(\frac{1}{y}) > \frac{1}{y} \Rightarrow \frac{1}{\exp(y)} = \exp(-\frac{1}{y}) < y$.

Hence, $\exp(-\frac{1}{y}) < y < \exp(y)$, so there is a c such that $\exp(c) = y$.

□

Definition. The function $\log : (0, \infty) \to \mathbb{R}$ is the inverse of $\exp : \mathbb{R} \to (0, \infty)$.

Proposition. For $x, y \in (0, \infty)$

- $\log(xy) = \log x + \log y$
- $\log(\frac{x}{y}) = \log x - \log y$

Proof.

- For the first proof

$$\begin{aligned} \exp(\log x + \log y) &= \exp(\log x)\exp(\log y) \\ &= xy \\ &= \exp(\log(xy)). \end{aligned}$$

  so by injectivity $\log xy = \log x + \log y$

- Because $\exp(0) = 1$ this means $\log 1 = 0$.

$$\begin{aligned} \log(\frac{y}{y}) &= \log y + \log(\frac{1}{y}) \\ 0 &= \log y + \log(\frac{1}{y}) \\ \log(\frac{1}{y}) &= -\log(y) \end{aligned}$$

  and the result follows.

□

Definition. For $a > 0$ and $x \in \mathbb{R}$

$$a^x = \exp(x \log a).$$

3 Derivatives

Definition. Suppose $S \subseteq \mathbb{R}$. A point $c \in S$ is called an **interior point** of S if there exists $\delta > 0$ such that

$$(c - \delta, c + \delta) \subseteq S.$$

Definition. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Suppose $c \in D$ is an interior point

of D. We say f is **differentiable** at c if

$$\lim_{h\to 0} \frac{f(c+h)-f(c)}{h} \quad \text{exists.}$$

The value $f'(c) := \lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$ is called **the derivative** of f at c.

If f is differentiable at every point $c \in D$, we say f is **differentiable** on D.

Proposition. Let $D \subseteq \mathbb{R}$ and let $c \in D$ be an interir point of D. If $f : D \to \mathbb{R}$ is differentiable at c, then it is continuous at c.

Proof. Suppose $\lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$ exists. Using a substition of $x = c + h$, this can be rewritten as

$$\begin{aligned}
\lim_{x\to c} \frac{f(x)-f(c)}{x-c} &= f'(c) \\
\lim_{x\to c} f(x) - f(c) &= \lim_{x\to c} (x-c)f'(c) \\
\lim_{x\to c} f(x) - f(c) &= 0
\end{aligned}$$

so f is continuous at c. $\square$

3.1 The Algebra of Derivatives

Theorem. Let $D \subseteq \mathbb{R}$ and let $c \in D$ be an interior point. Suppose that $f, g : D \to \mathbb{R}$ are differentiable at c. Then

- $f + g$ is differentiable at c with
$$(f+g)'(c) = f'(c) + g'(c).$$
- $f - g$ is differentiable at c with
$$(f-g)'(c) = f'(c) - g'(c).$$
- αf is differentiable at c with $(\alpha f)'(c) = \alpha f'(c)$
- (Product rule) fg is differentiable at c with
$$(fg)'(c) = f'(c)g(c) - f(c)g'(c).$$
- (Quotient rule). If $g(c) \neq 0$ then $\frac{f}{g}$ is differentiable at c with
$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Proof. The first three points can be proven relatively easily using the al-

gebra of function limits.

- First, it is easy to show that

$$(fg)(c+h)-(fg)(c) = (f(c+h)-f(c))\,g(c+h)+f(c)\,(g(c+h)-g(c))\,.$$

Then

$$\begin{aligned}(fg)'(c) &= \lim_{h\to 0}\frac{(f(c+h)-f(c))\,g(c+h)+f(c)\,(g(c+h)-g(c))}{h}\\ &= f'(c)g(c)+f(c)g'(c).\end{aligned}$$

- We only need to prove that $\left(\frac{1}{g}\right)'(c) = \frac{-g'(c)}{(g(c))^2}$, and can then use the product rule. Using the continuity of g with $\epsilon = \frac{|g(c)|}{2}$, we can find a δ such that for all $h \in (c-\delta, c+\delta)$, $g(c+h) \neq 0$, so it is well defined. Then

$$\begin{aligned}\lim_{h\to 0}\frac{(\frac{1}{g})(c+h)-(\frac{1}{g})(c)}{h} &= \lim_{h\to 0}\frac{\frac{1}{g(c+h)}-\frac{1}{g(c)}}{h}\\ &= -\lim_{h\to 0}\frac{g(c+h)-g(c)}{h}\cdot\frac{1}{g(c+h)g(c)}\\ &= -\frac{g'(c)}{g(c)^2}.\end{aligned}$$

□

3.2 Derivatives of Basic Functions

Proposition. Let $f : D \to \mathbb{R}, f(x) = x^2$ where $D = \mathbb{R}$ if $n > 0$ and $\mathbb{R} \setminus \{0\}$ otherwise. Then f is differentiable on D with derivative

$$f'(x) = nx^{n-1}.$$

Proof.

- Suppose $n \geq 0$. Then using induction on n, which is true $n = 0$, then assuming true for $n = k$, for $n = k+1$

$$\begin{aligned}f'(x) &= \frac{d}{dx}(x)(x)^k\\ &= x^k + (k)x^{k-1}x\\ &= (k+1)x^k.\end{aligned}$$

- Suppose $n < 0$. Then $f = \frac{1}{g}$ where $g(x) = x^{-n}$. Then using the quotient rule

$$\begin{aligned}&= -\frac{g'(x)}{g(x)^2}\\ &= nx^{n-1}.\end{aligned}$$

□

Corollary. Let $p : \mathbb{R} \to \mathbb{R}$ be a polynomial with

$$P(x) = a_n x^n + ... + a_1 x + a_0.$$

Then P is differentiable on $\mathbb{R}$ and

$$p'(x) = n a_n x^{n-1} + ... + a_1.$$

Proposition. The functions $\exp, \cos, \sin$ are differentiable with

- $\exp'(x) = \exp(x)$
- $\cos'(x) = -\sin(x)$
- $\sin'(x) = \cos(x)$

Proof. The arguments for all three are similar, so let the desired function be $f : \mathbb{R} \to \mathbb{R}$.

- Step 1: Show f is differentiable at 0 and find $f'(0)$.
- Step 2: In the derivative of $f(x)$, write $f(x+h)$ as a product of $f(h)$, then as this is already known it can be evaluated.

□

3.3 The Chain Rule

Theorem. Let $D, E \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$ and $g : E \to D$. Let $c \in E$ be an interior point, such that $g(c)$ is an interior point of D. If

- g is differentiable at c
- f is differentiable at $g(c)$
-

then $f \circ g$ is differentiable at c with

$$(f \circ g)'(c) = f'(g(c))g'(c).$$

Proof. Let $h \in \mathbb{R} \setminus \{0\}$ be such that $c + h \in E$. If $g(c+h) \neq g(c)$ then

$$\begin{aligned}\frac{(f \circ g)(c+h) - (f \circ g)(c)}{h} &= \frac{f(g(c+h)) - f(g(c))}{h} \\ &= \frac{f(g(c+h)) - f(g(c))}{g(c+h) - g(c)} \cdot \frac{g(c+h) - g(c)}{h}.\end{aligned}$$

We need to define $q : D \to \mathbb{R}$ by

$$q(y) = \begin{cases} \frac{f(y)-f(g(c))}{y-g(c)} & y \in D \setminus \{g(c)\} \\ f'(g(c)) & y = g(c) \end{cases}$$

which lets us more easily prove properties needed. Then,

$$\frac{(f \circ g)(c+h) - (f \circ g)(c)}{h} = q(g(c+h)) \cdot \frac{g(c+h) - g(c)}{h}$$

which is true for both cases of q, letting us avoid the case $g(c+h) = g(c)$. Then

$$\lim_{y \to g(c)} q(y) = f'(g(c)) = q(g(c))$$

so q is continuous, and so is $q \circ g$, therefore

$$\begin{aligned} (f \circ g)'(c) &= \lim_{h \to 0} q(g(c+h)) \lim_{h \to 0} \frac{g(c+h) - g(c)}{h} \\ &= \lim_{h \to 0} q(g(c+h))g'(c) \\ &= f'(g(c))g'(c). \end{aligned}$$

□

3.4 The derivative of inverse functions

Theorem. Let $D, E \subseteq \mathbb{R}$ and let $f : D \to E$ bijective function. Suppose $c \in D$ and $f(c) \in E$ are interior points. If f is differentiable at c with $f'(c) \neq 0$ then f^{-1} is differentiable at $f(c)$ with

$$(f^{-1})'(f(c)) = \frac{1}{f'(c)}.$$

Proof. Non examinable. □

4 Working With Derivatives

4.1 Extremal Points

Definition.

- $c \in D$ is called a **local maximum point** of $f : D \to \mathbb{R}$ if there exists $\delta > 0$ such that for all $x \in D$ with $|x - c| < \delta$ we have

$$f(x) \leq f(c).$$

- $c \in D$ is called a **local minimum point** of $f : D \to \mathbb{R}$ if there exists a $\delta > 0$ such that for all $x \in D$ with $|x - c| < \delta$ we have

$$f(c) \leq f(x).$$

Theorem. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Suppose

- c is an interior point
- c is either a local maximum or a local minimum
- f is differentiable at c.

Then
$$f'(c) = 0.$$

Proof. Suppose c is a local maximum. Using this and that c is an interior point we can find δ such that
$$f(x) \leq f(c) \quad \forall\, x \in (c-\delta, c+\delta).$$
Then for all $n > \frac{1}{\delta}$ we have
$$f(c) \geq f(c - \frac{1}{n}), \quad f(c) \geq f(c + \frac{1}{n})$$
so
$$\frac{f(c+\frac{1}{n}) - f(c)}{\frac{1}{n}} \leq 0 \leq \frac{f(c-\frac{1}{n}) - f(c)}{-\frac{1}{n}}.$$
Taking the limit to infinity then gives that $f'(c) = 0$ by the sandwich theorem. □

4.2 The Mean Value Theorem

Theorem. (Rolle's Theorem)
Let $a < b$ and $f : [a, b] \to \mathbb{R}$. Suppose f is continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$ then there exists $c \in (a, b)$ with $f'(c) = 0$.

Proof. By the Weierstrass sextreme value theorem f has a global minimum $c_1 \in [a, b]$ and a global maximum $c_2 \in [a, b]$. If either of these are in (a, b) we are done. Otherwise the minimum and maximum are equal so for any x,
$$f(a) \leq f(x) \leq f(x) \quad \Rightarrow \quad f(x) = f(a)$$
and $f'(x) = 0$. □

Theorem. (Mean Value Theorem)
Let $a < b$ and $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Let $g(x) := f(x) - x\frac{f(b)-f(a)}{b-a}$. Then

$$\begin{aligned} g(b) - g(a) &= f(b) - f(a) - b\frac{f(b)-f(a)}{b-a} + a\frac{f(b)-f(a)}{b-a} \\ &= f(b) - f(a) - \frac{(b-a)(f(b)-f(a))}{b-a} \\ &= 0. \end{aligned}$$

Thus we can use Rolle's theorem to find c with $g'(c) = 0$, which can be rearanged for the desired result. □

Corollary. Let $a < b$ and $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $f'(x) = 0$ for all $x \in (a, b)$ then f is constant on $[a, b]$.

Proof. For $x \in (a, b]$, by the mean value theorem there exists $c \in (a, x)$ such that

$$f'(c) = 0 = \frac{f(x) - f(a)}{x - a} \quad \Rightarrow f(x) = f(a).$$

□

Theorem. Let $a < b$, and $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$(f(b) - f(a))(g'(c)) = (g(b) - g(a))f'(c)$$

or alternatively that

$$\frac{f'(c)}{g'(c)} = \frac{f(c) - f(a)}{g(b) - g(a)}.$$

Proof. Consider $h : [a, b] \to \mathbb{R}$ with

$$h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x).$$

Then it can be shown that

$$h(a) = f(b)g(a) - f(a)g(b) = h(b).$$

Then as h is differentiable we can use Rolle's theorem to find a point c where $h'(c) = 0$ which can be rearanged for the desired result. □

4.3 Monotonicity

Definition. Let $D \subseteq \mathbb{R}$ be an interval and $f : D \to \mathbb{R}$.

- f is **increasing** on D if

$$x < y \Rightarrow f(x) \leq f(y).$$

- f is **decreasing** D if

$$x < y \Rightarrow f(x) \geq f(y).$$

- f is strictly **increasing** on D if

$$x < y \Rightarrow f(x) < f(y).$$

- f is **decreasing** D if

$$x < y \Rightarrow f(x) < f(y).$$

If any of these conditions are satisfied we say f is **monotone**.

Remark. If $f : D \to \mathbb{R}$ is strictly increasing or decreasing, then it follows that $f(x) \neq f(y)$ if $x \neq y$. Hence this function is injectvie.

Theorem. Let $a < b$ and $f : [a, b] \to \mathbb{R}$. Suppose f is continuous on $[a, b]$ and differentiable on (a, b).

- f is increasing in $[a, b]$ if and only if $f'(x) \geq 0$ for all $x \in (a, b)$.
- f is decreasing in $[a, b]$ if and only if $f'(x) \leq 0$ for all $x \in (a, b)$
- f is strictly increasing in $[a, b]$ if $f'(x) > 0$ for all $x \in (a, b)$
- f is strictly decreasing in $[a, b]$ if $f'(x) < 0$ for all $x \in (a, b)$

Proof. We only prove the first statement.
($\Rightarrow$) Suppose $x \in (a, b)$ and consider $(x_n)_{n\in\mathbb{N}} \to x$ with $x_n = x + \frac{1}{n}$ for n large enough so that $x_n \in (a, b)$. Then as f is increasing

$$\frac{f(x + \frac{1}{n}) - f(x))}{\frac{1}{n}} \geq 0$$

thus taking the limit gives that $f'(x) \geq 0$.
($\Leftarrow$) Consider $x_2 > x_1 \in (a, b)$. Then by the mean value theorem there exists c such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1) \geq 0.$$

$\square$

Corollary. Let c be an interior point of $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$ differentiable on $(c - r) \cup (c, c + r)$ for $r > 0$.

- If $f'(x) \geq 0$ for $x \in (c - r)$ and $f'(x) \leq 0$ for $x \in (c + r)$, c is a local maximum of f.

- If $f'(x) \leq 0$ for $x \in (c-r)$ and $f'(x) \geq 0$ for $x \in (c+r)$, c is a local minimum of f.

Proof. For the first statement, if $x \in [c - \frac{r}{n}, c)$, then f is increasing on $[x, c]$ so $f(x) \leq f(c)$, and if $x \in (c, c + \frac{r}{2}]$ then f is decreasing on $[c, x]$ so $f(x) \leq f(c)$. If $x = c$ then clearly $f(x) \leq f(c)$. Thus c is a maximum. $\square$

4.4 L'Hôpital's Rule

Theorem. Suppose $f, g : (a, b) \setminus \{c\} \to \mathbb{R}$ are differentiable and satisfy

$$\lim_{x \to c} f(x) = 0 \quad \text{and} \quad \lim_{x \to c} g(x) = 0$$

but $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a, b) \setminus \{c\}$.
If $\lim\limits_{x \to c} \frac{f'(x)}{g'(x)}$ is defined then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}.$$

Proof. We extend f and g to (a, b) by setting $f(c) = 0 = g(c)$. Therefore they are continuous on (a, b). As

$$\lim_{x \to c} \frac{f'(x)}{g'(x)} = L \quad \text{and } c \in (a, b)$$

we see that for $\epsilon > 0$ there exists a $\delta > 0$ where $(c - \delta, c + \delta) \subseteq (a, b)$ and

$$0 < |z - c| < \delta \Rightarrow |\frac{f'(z)}{g'(z)} - L| < \epsilon.$$

Now, let $0 < |x - c| < \delta$.

- If $x > c$, apply the Cauchy mean value theorem on $[c, x]$. Then there exists a $y \in (c, x)$ such that

$$\frac{f'(y)}{g'(y)} = \frac{f(x) - f(c)}{g(x) - g(c)} = \frac{f(x)}{g(x)}.$$

- If $x < c$, then the Cauchy mean value theorem says there exists $y \in (x, y)$ such that

$$\frac{f'(y)}{g'(y)} = \frac{f(x)}{g(x)}.$$

In both cases, $0 < |y - c| < |x - c| < \delta$, thus

$$\left| \frac{f(x)}{g(x)} - L \right| = \left| \frac{f'(y)}{g'(y)} - L \right| < \epsilon.$$

$\square$